Artificial Intelligence Solution Implementation

Quantity Forecasting Feasibility Study: LC-CLS Startup Kit

1. Business Context

Labcorp Central Laboratory Services (LC-CLS) operates as a central laboratory provider supporting pharmaceutical clinical trials across more than 100 countries, coordinating specimen processing, logistics, and kit supply globally. Start-up kit allocation (predicting and dispatching visit-specific sample collection materials before first patient enrolment) sits between two inescapable operational errors: excess inventory expires under the six-month shelf-life constraint governing biological consumables, and insufficient supply disrupts patient visits with knock-on regulatory consequences under Good Clinical Practice requirements (Getz, Smith and Kravet, 2023). Each failure carries a recovery cost; wasted cold-chain logistics in the first case, delayed patient visits and potential regulatory breach in the second.

Kit supply optimisation was identified in the Unit 9 LC-CLS AI proposal as a priority use case, recommending a gradient boosting model with a quantile regression objective to predict per-site, per-visit-type kit quantities from site-level and study-level features. Among the use cases Unit 9 identified, kit quantity prediction was selected here because its outcome is directly measurable, its failure mode is patient-facing, and it yields a continuous target suitable for empirical evaluation (Studer et al., 2021). This report implements a prototype regression experiment on publicly available healthcare supply chain data to establish an empirical feasibility case before any production commitment is made.

2. Dataset Justification

Selection fell on the USAID Supply Chain Management System (SCMS) Delivery History, comprising 10,324 healthcare commodity shipments procured under PEPFAR across 43 countries between 2006 and 2015 (USAID, 2015). Three grounds justified the choice. The data are genuinely healthcare-domain, covering antiretroviral drugs (ARV), HIV rapid diagnostic test kits (HRDT), and antimalarial treatments (ACT); these commodity categories proxy clinical trial kit types given their shared characteristics as regulated healthcare consumables. Feature coverage maps onto LC-CLS variables: country-to-site region, product group-to-therapeutic area, and shipment mode-to-logistics-urgency tier. The dataset is open access, supporting reproducibility, but does not replicate the LC-CLS regulatory constraints discussed in Section 6. After cleaning, WEKA confirmed 12 attributes and 10,306 instances with zero missing values.

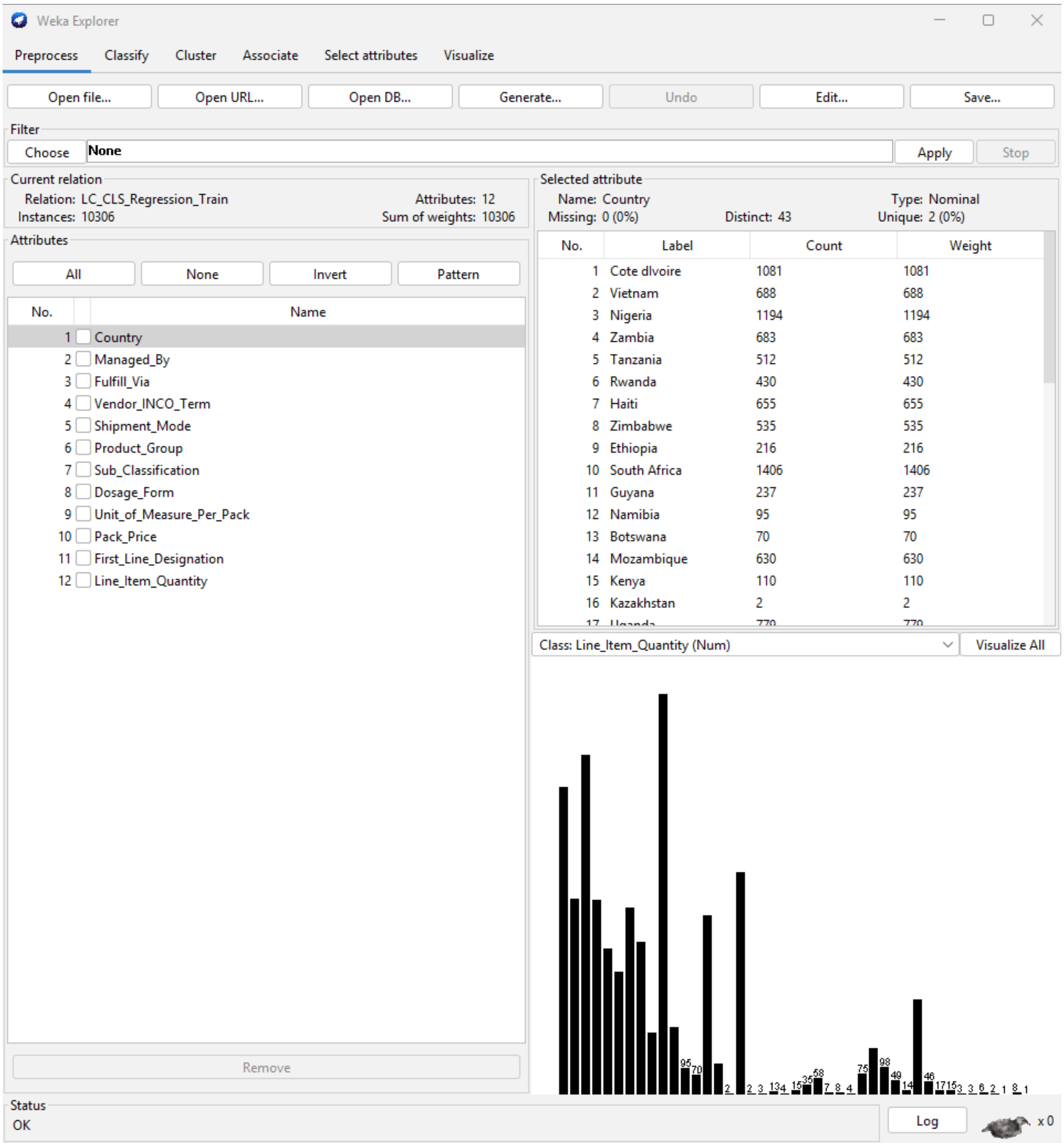

Figure 1: *Preprocess tab: 12 attributes, 10,306 instances.*

The nine categorical attributes (Appendix, Figures A1-A8, A11) show expected concentration patterns: ARV dominates Product_Group at 82.9%, Air dominates Shipment_Mode at 62.7%, and PMO-US accounts for 99.4% of Managed_By, contributing almost no discriminative signal.

Distributional inspection of the two numeric predictors (Appendix, Figures A9-A10) confirms that Pack_Price is substantially right-skewed (mean 21.95, sd 45.64), reflecting the price differential between generic antiretrovirals and specialised biologics. Unit_of_Measure_Per_Pack ranges from 1 to 1,000 units per pack (mean 77.9, sd 76.6), concentrated at standard dispensing sizes. First_Line_Designation is binary, with 68% flagged as first-line (Appendix, Figure A11). No outlier removal was applied; extreme values are genuine procurement events (Getz, Smith and Kravet, 2023).

Line_Item_Quantity was used directly as a continuous regression target without transformation. It ranges from 1 to 619,999 units (mean 18,361, standard deviation 40,064) and is substantially right-skewed (Figure 2).

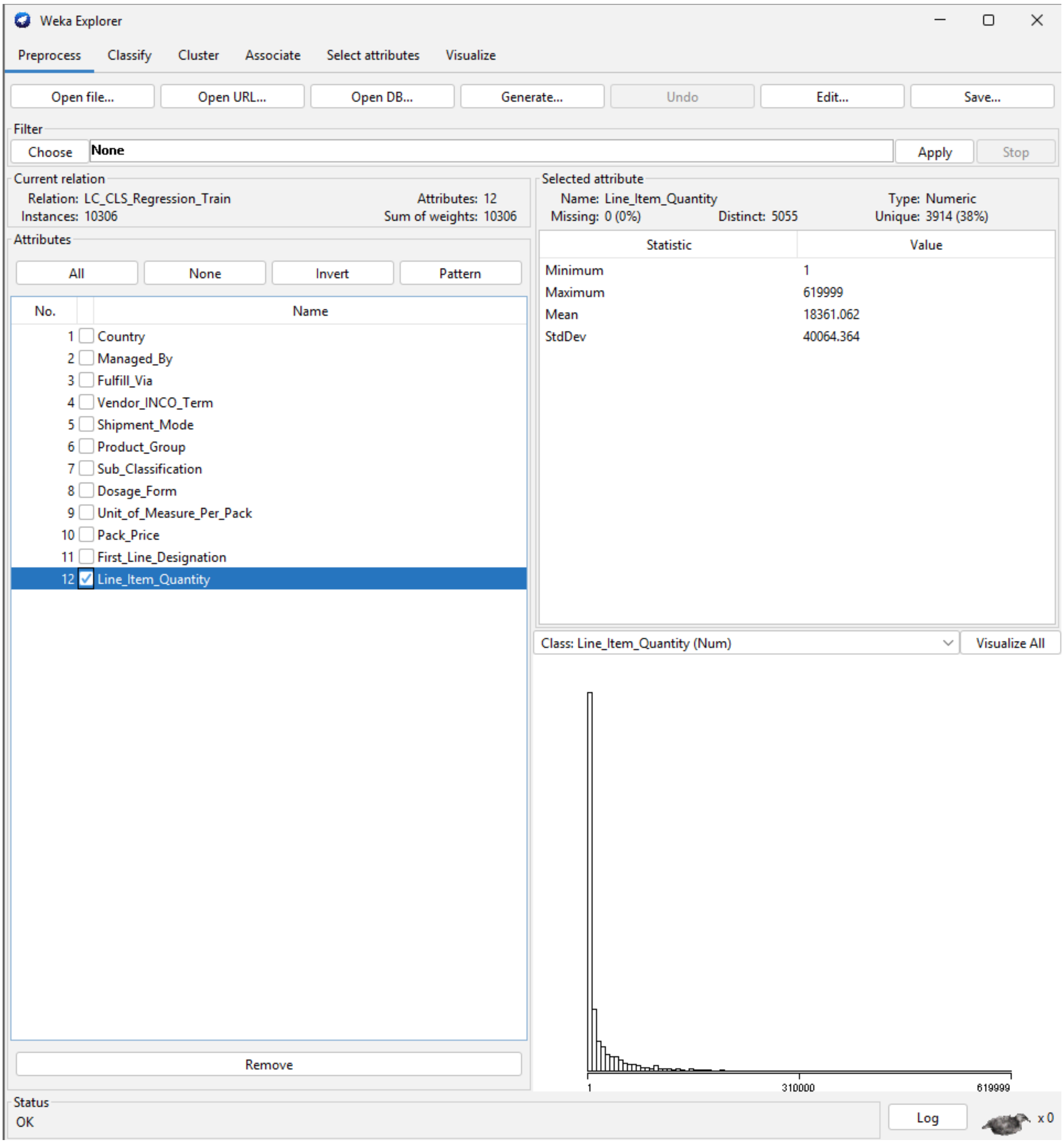

Figure 2: *Line_Item_Quantity distribution.*

Preprocessing reduced the original 33-column dataset to 12 analytical attributes with zero missing values across all 10,306 retained rows: nine categorical predictors and two numeric predictors (pack price and unit of measure per pack), with Line_Item_Quantity as the regression target. Twenty-one columns were removed: administrative identifiers carry no predictive signal; date fields introduce leakage risk across folds; high-cardinality free-text fields are captured at a more stable abstraction by retained categorical attributes. Two leaky features were removed: Line Item Value is derivable from Line Item Quantity and Pack Price by division; Line Item Insurance encodes the same signal (Kapoor and Narayanan, 2023). Four post-dispatch numerics were excluded: Unit Price is derived from retained predictors; Weight, Freight Cost, and Manufacturing Site reflect outcomes rather than drivers of quantity. Eighteen rows where the Pack Price equalled zero were excluded. Feature exclusion prioritised leakage avoidance and causal relevance; this may discard latent predictive signal, a trade-off examined in Section 6.

3. Modelling Approach

The experiment adopts a supervised learning paradigm, training on labelled historical records to predict a continuous quantity target; regression was chosen because the business question is how many kits, not which category. Following CRISP-ML(Q), it progressed through data understanding, preprocessing, modelling, and evaluation phases with quality gates at each transition (Studer et al., 2021). Regression was performed in WEKA 3.8.6.

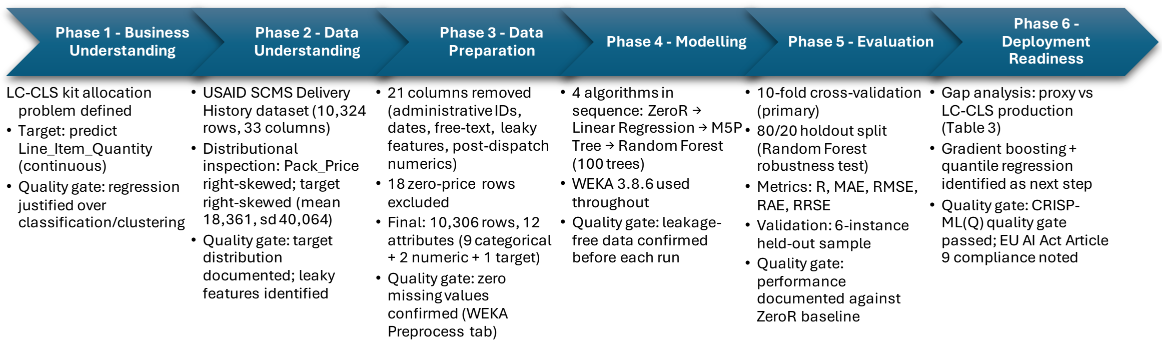

***Figure 3**: CRISP-ML(Q) pipeline (Studer et al., 2021).*

Data understanding confirmed Line_Item_Quantity as the continuous regression target. The right-skewed target distribution (Figure 2) means RMSE will be disproportionately influenced by large-volume misses, a procurement characteristic, not a data quality issue, reported explicitly for each model.

Model evaluation employed stratified 10-fold cross-validation throughout, preserving distributional consistency across folds and limiting the influence of fold-level sampling artefacts on reported performance (Studer et al., 2021). Ten folds were chosen as computationally practical and more representative than any single train/test split. A supplementary 80/20 percentage split was applied to Random Forest as a robustness test on 2,061 unseen instances. Primary metrics were Correlation coefficient (R), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Relative Absolute Error (RAE), and Root Relative Squared Error (RRSE). RMSE weights large misses more heavily by squaring each error, whilst MAE treats all errors proportionally; Wang (2022) argues neither metric is inherently superior, so both are reported here to reflect the right-skewed error distribution.

4. Algorithm Rationale

Model selection reflects the dataset: categorical features with a right-skewed target favour tree-based approaches; the baseline-to-ensemble sequence satisfies CRISP-ML(Q)'s incremental complexity requirement (Studer et al., 2021). This progression follows the structure that the LC-CLS AI Framework expects of a feasibility experiment and aligns with the risk management approach that EU AI Act Article 9 requires before deploying high-risk AI systems (Regulation (EU) 2024/1689, 2024).

Scenario analysis clarifies why this sequence was chosen over a single-model approach, a choice the interpretability literature frames as context-dependent rather than universally optimal (Rudin et al., 2022). Where a regulatory audit requires full traceability of every prediction to specific feature contributions, Linear Regression is the only viable option; each coefficient is directly readable. Where operational decision-support requires that a supply manager challenge allocation rules without statistical expertise, M5P provides auditable node-level split logic. Random Forest establishes how much predictive performance these features can realistically yield before investing in production.

ZeroR is the starting point: it ignores all features and simply predicts the average training quantity (18,361 units) for every shipment. Figure 4 confirms these figures; the near-zero correlation coefficient (-0.029) confirms no meaningful relationship between the prediction and actual order volumes, as expected.

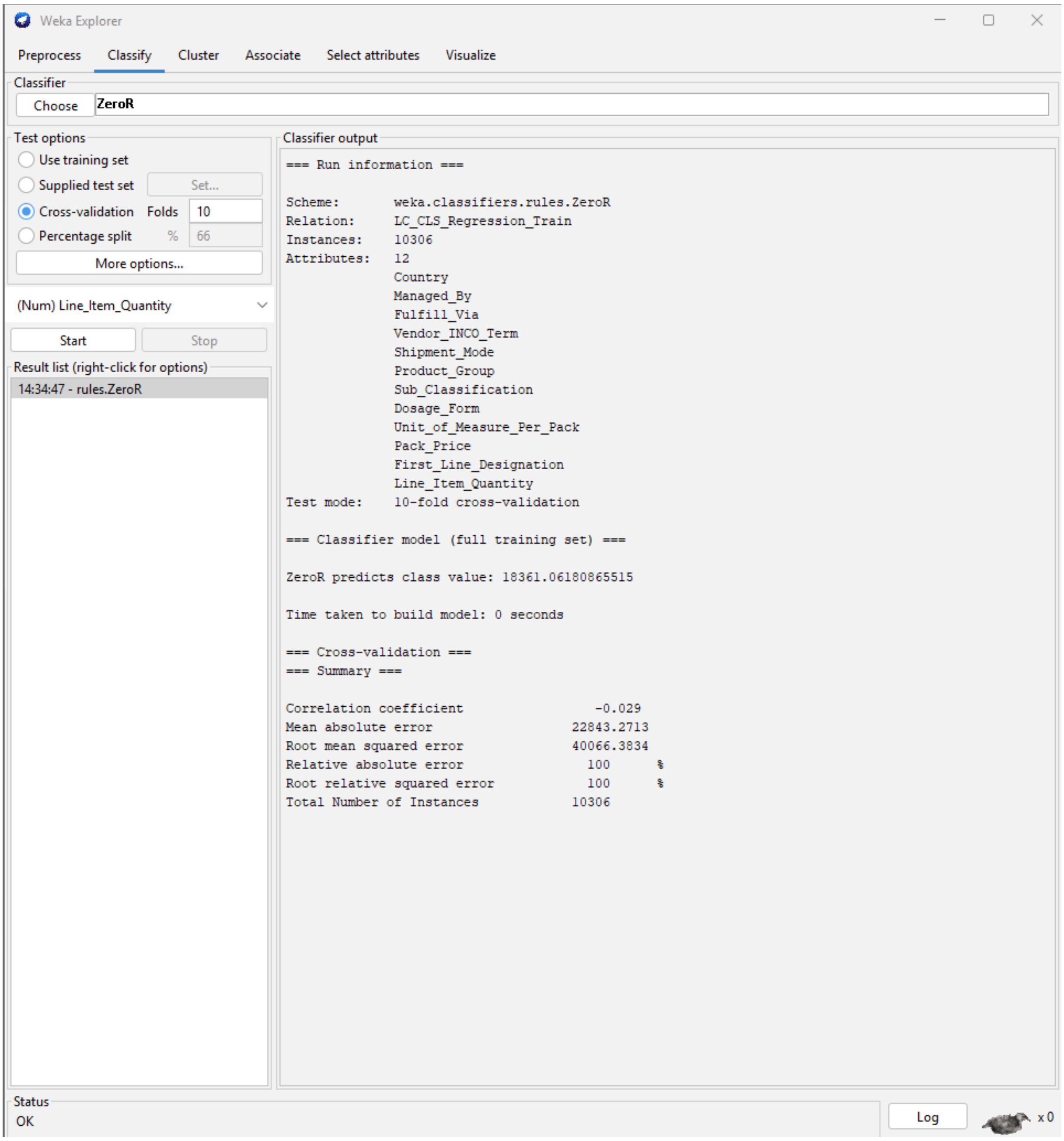

Figure 4: *ZeroR output.*

Linear Regression was included because its results are fully transparent. Each predictor gets a numerical coefficient readable directly from Figure 5: Ocean shipments carry the largest increase (+32,588 units); higher Pack_Price is associated with lower quantities (-17.18 per unit); and First_Line_Designation adds +4,477 units, consistent with first-line antiretrovirals being purchased at scale. This satisfies EU AI Act Article 9's requirement that high-risk AI decisions be explainable and auditable (Regulation (EU) 2024/1689, 2024).

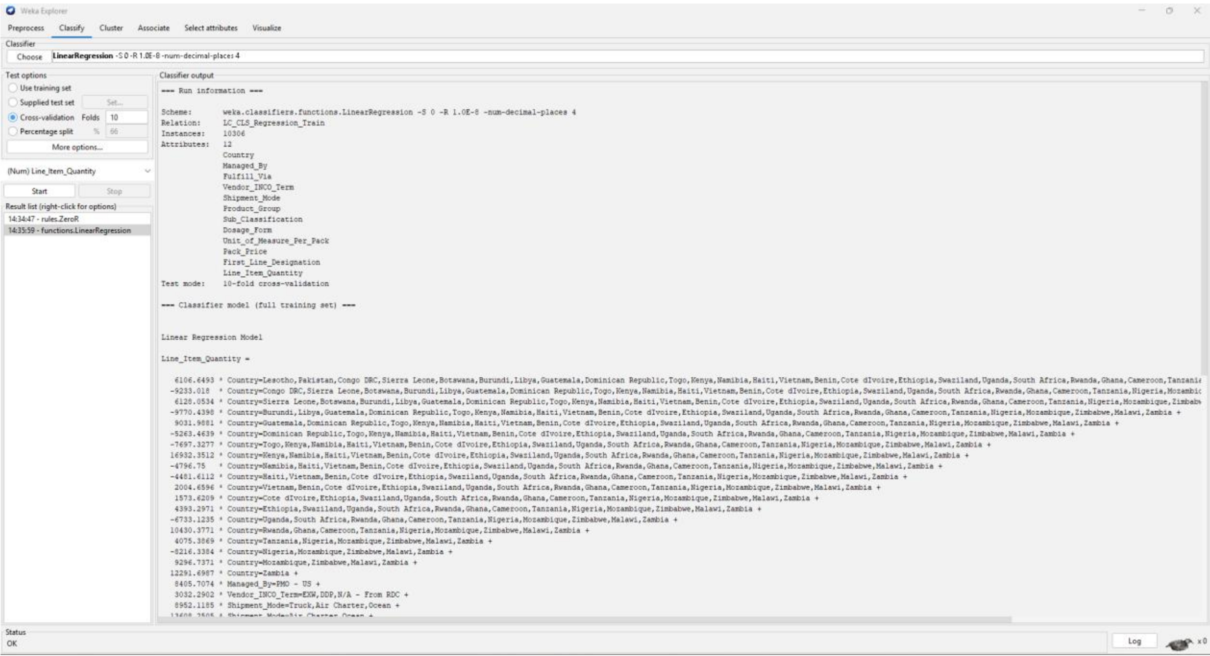

Figure 5: Linear Regression coefficients.

M5P, WEKA's pruned model tree (University of Waikato, 2024), produces human-readable allocation rules with linear regression models at each leaf. The root split on Dosage_Form followed by Pack_Price <= 3.225 reveals a coherent commercial logic: tablet formulations at low pack prices are the high-volume ARV tier. With 80 rules, M5P is more expressive than Linear Regression whilst remaining auditable; any leaf prediction can be challenged, satisfying the accountability standard that Rudin et al. (2022) require in consequential settings.

Random Forest combines 100 decision trees, outperforming any single tree (Breiman, 2001) and setting a benchmark for the Unit 9 production model. CRISP-ML(Q) requires this benchmark to be established on clean data before moving forward (Studer et al., 2021; Regulation (EU) 2024/1689, 2024).

5. Results and Output Analysis

Both MAE and RMSE confirm a predictive signal above the baseline. The right-skew of Line_Item_Quantity is the structural obstacle for all models, consistent with comparable healthcare supply chain studies (Ganesh and Dass, 2025). Random Forest achieves a 43.0% reduction in MAE and a 23.7% reduction in RMSE over ZeroR, confirming a genuine supply-quantity signal in the feature set.

Table 1: *Regressor performance, 10-fold CV and 80/20 holdout.*

Model	R	MAE	RMSE	RAE	RRSE	MAE improvement
ZeroR (baseline)	-0.029	22,843	40,066	100.0%	100.0%	-
Linear Regression	0.493	18,376	34,855	80.4%	87.0%	-19.6%
M5P Tree (CV)	0.621	14,293	31,427	62.6%	78.4%	-37.4%
Random Forest (CV)	0.660	13,011	30,588	57.0%	76.3%	**-43.0%**
Random Forest (80/20)	0.629	13,347	34,155	57.7%	78.1%	-41.6%

Linear Regression achieves R = 0.493 and MAE = 18,376, a 19.6% improvement over ZeroR. Figure 5 shows the coefficient list, confirming directionally coherent results; Figure 6 shows the wide residuals, reflecting the model's inability to capture non-linear interactions in a predominantly categorical feature set. M5P achieves R = 0.621, MAE 14,293 (37.4% improvement).

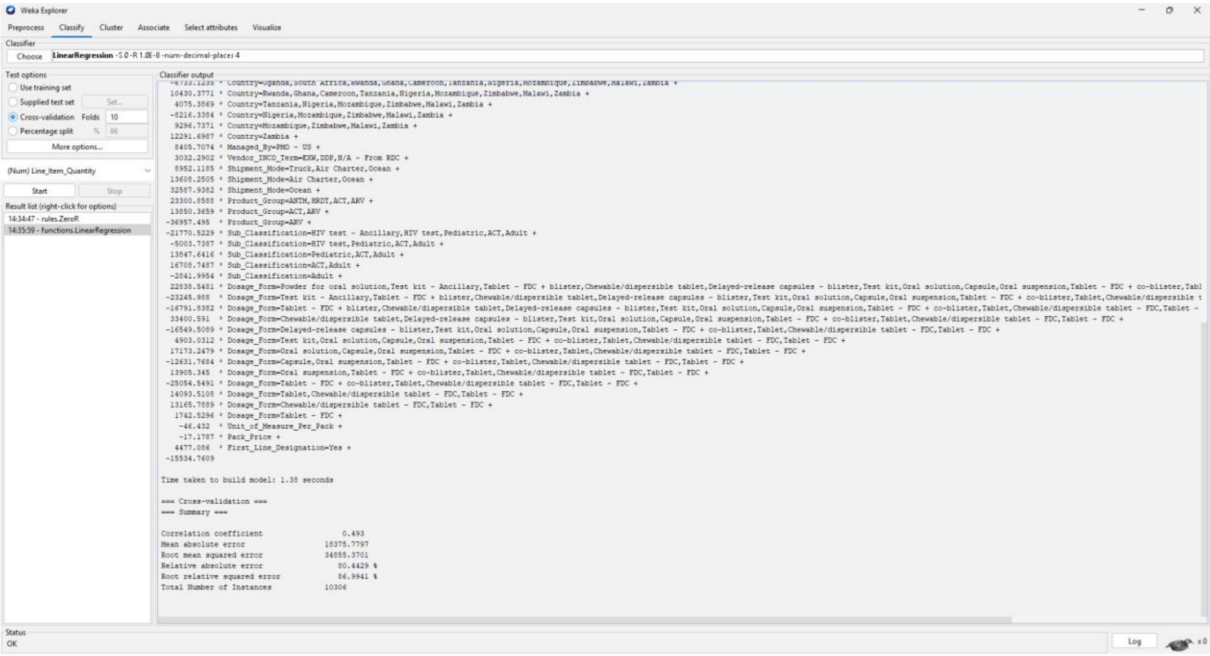

Figure 6: *Linear Regression summary.*

Figure 7 shows the tree structure: the root splits on Dosage_Form separates tablet formulations immediately, then Pack_Price <= 3.225 distinguishes the high-volume generic ARV tier. This split is economically coherent given that first-line antiretrovirals dominate PEPFAR procurement.

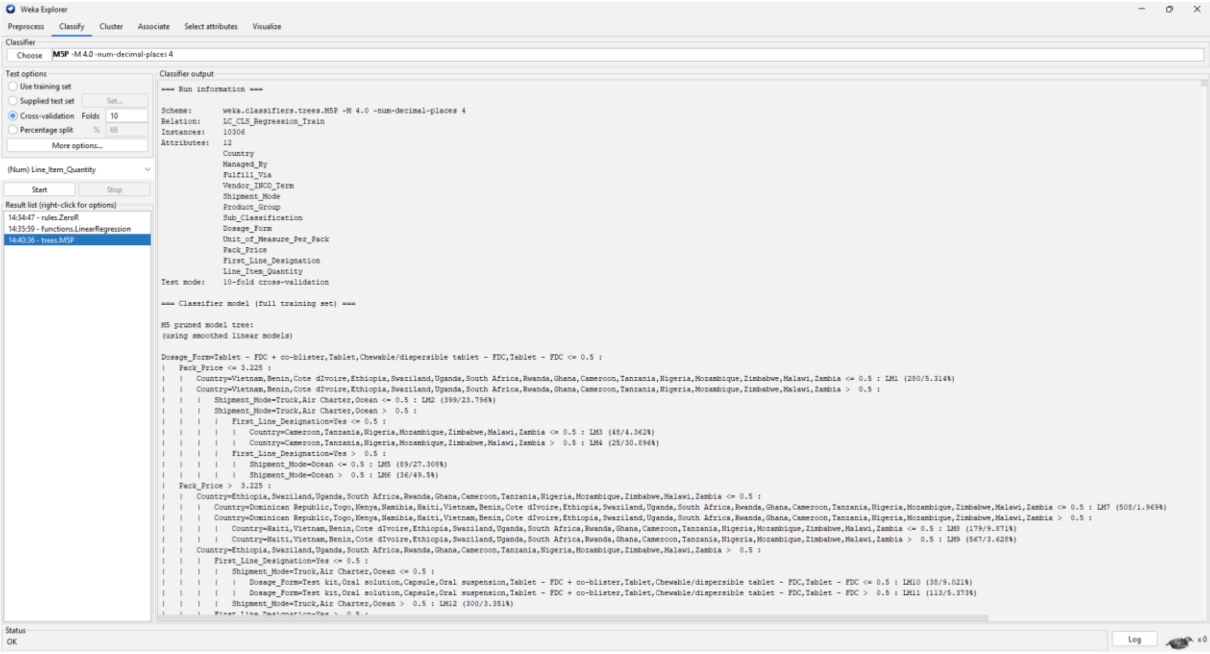

Figure 7: *M5P tree structure.*

Figure 8 confirms the performance gain, with Relative Absolute Error falling from 80.4% to 62.6%. Random Forest achieves the best cross-validated performance at R = 0.660, MAE 13,011, RMSE 30,588.

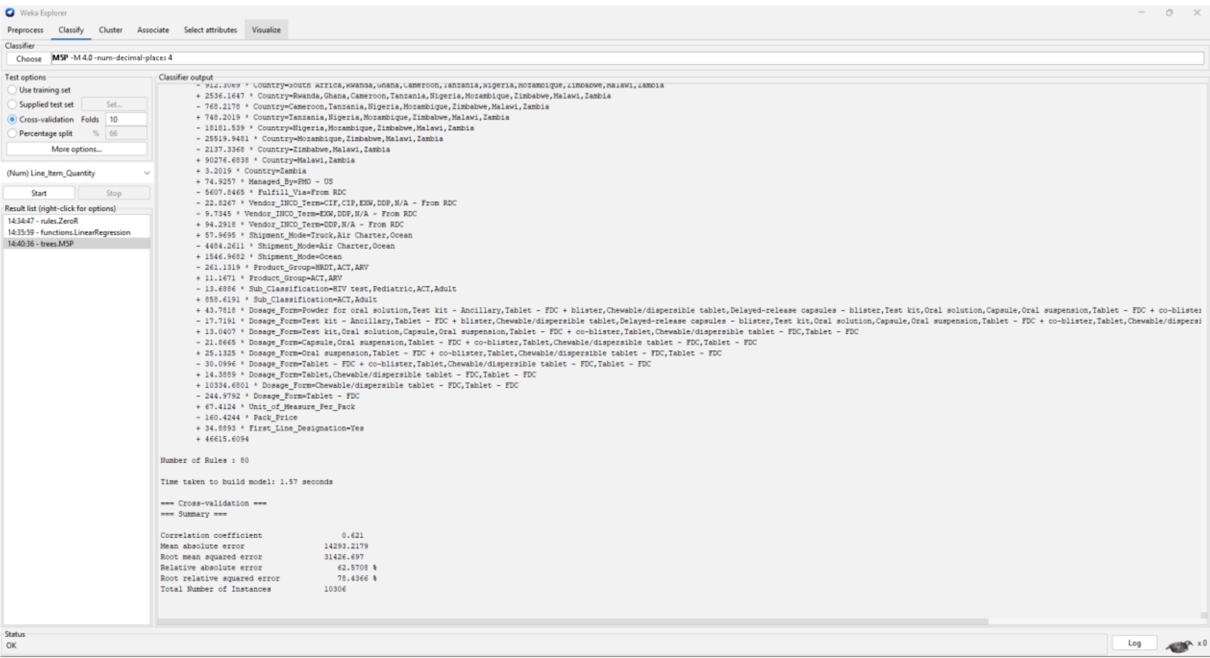

Figure 8: *M5P summary.*

Figure 9 shows the ensemble output; at 57.0% RAE, the reduction from M5P's 62.6% confirms that combining 100 trees reduces variance.

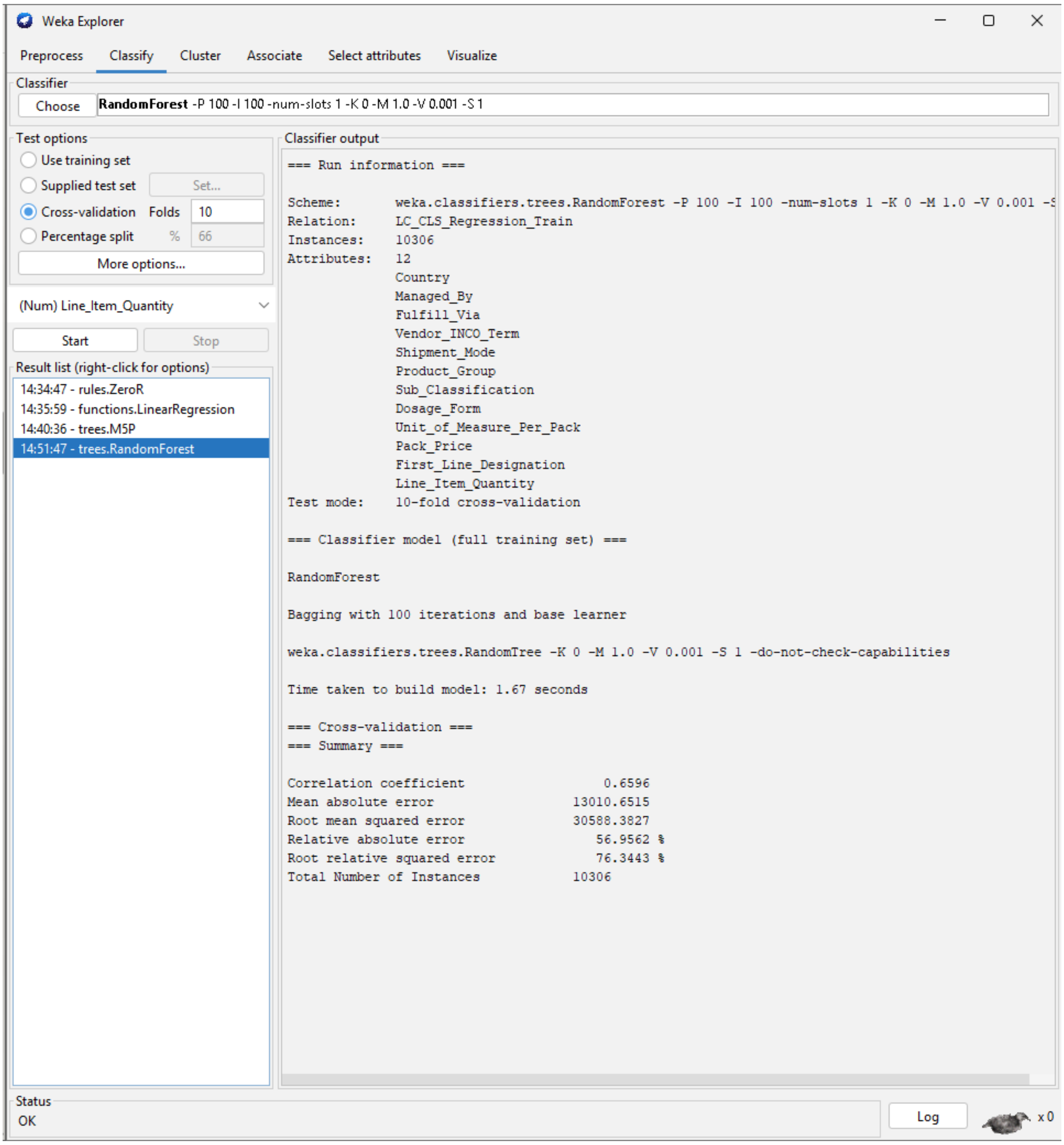

Figure 9: *Random Forest, 10-fold CV.*

The 80/20 split yields slightly weaker results: R = 0.629 and RMSE = 34,155, likely because the unshuffled final 20% concentrates on a procurement period unseen during training; cross-validation figures are the more trustworthy performance estimate.

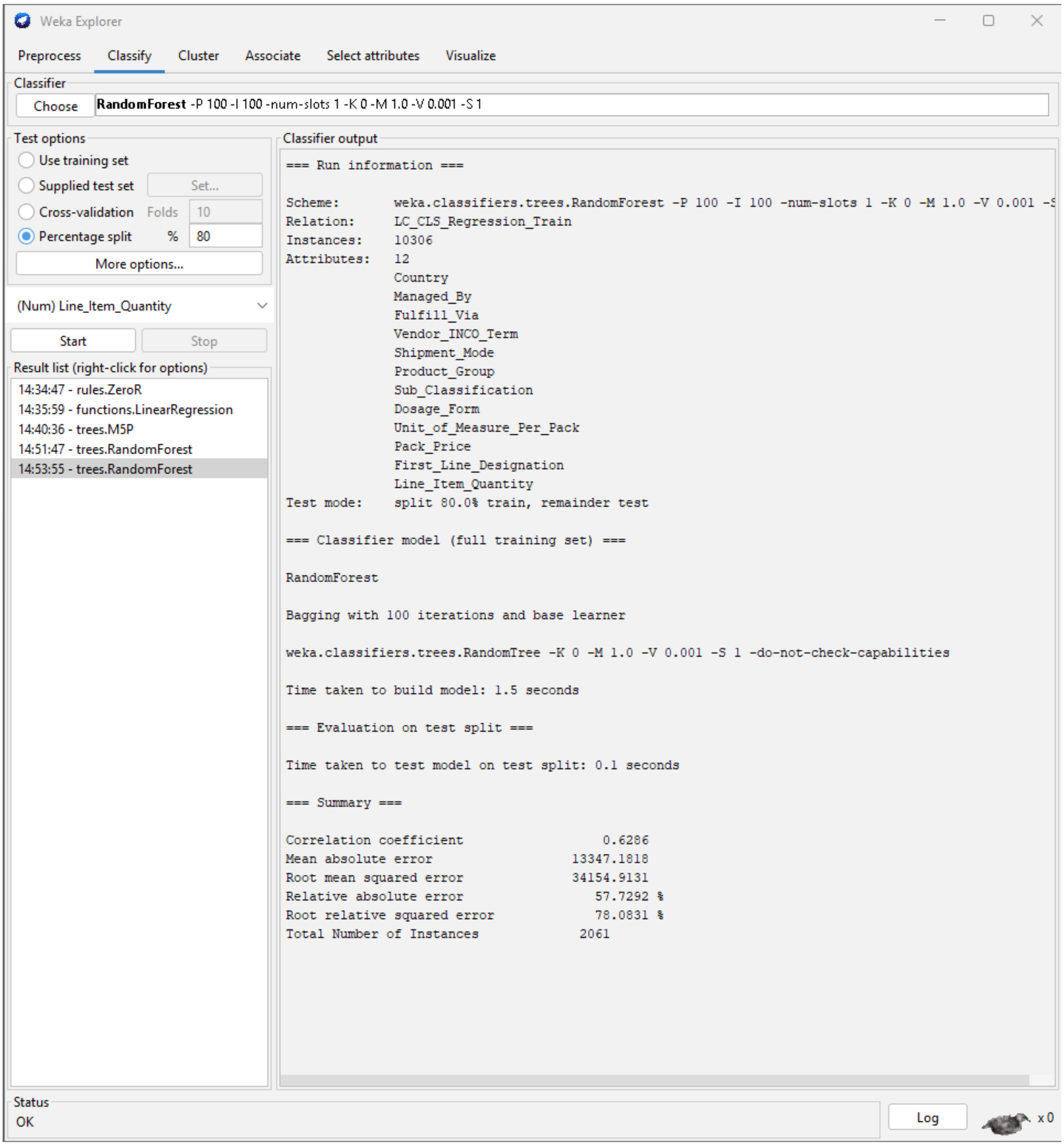

Figure 10: *Random Forest, 80/20 split.*

The larger performance gain occurs between Linear Regression and M5P (R: 0.493 to 0.621) than between M5P and Random Forest (R: 0.621 to 0.660), suggesting that non-linear categorical interaction is the primary predictive obstacle, not ensemble variance. The high-volume tail would remain unreliable under any RMSE-optimised approach on this distribution; that is structural, not algorithmic.

Table 2: *Six-instance validation sample, Random Forest CV (Figures 11-12).*

Country	Product	Mode	Actual LIQ	Predicted LIQ	Error	%
South Africa	ARV	Truck	42	251	+209	+497.6%
Mozambique	HRDT	Ocean	7,641	6,969	-672	-8.8%
Rwanda	ARV	Ocean	73,195	110,813	+37,618	+51.4%
Nigeria	ANTM	Air	300	2,314	+2,014	+671.2%
Kenya	HRDT	Truck	1,550	2,579	+1,029	+66.4%
Côte d'Ivoire	ACT	Air	25,000	31,768	+6,768	+27.1%

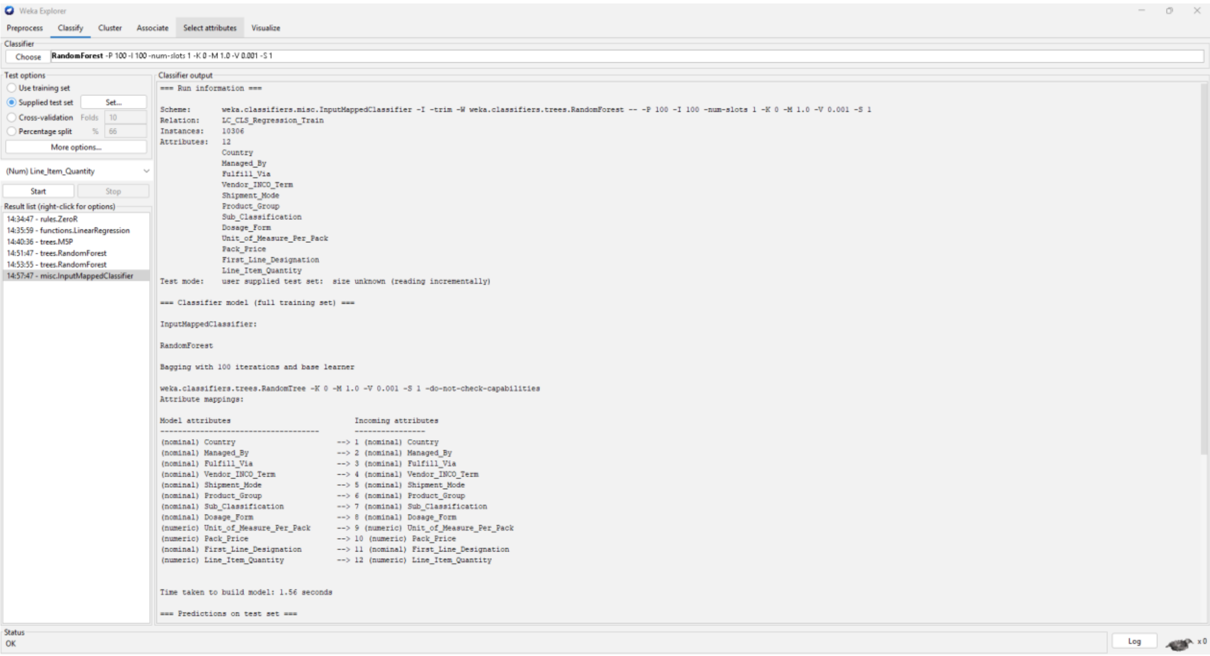

Figure 11: *Validation setup.*

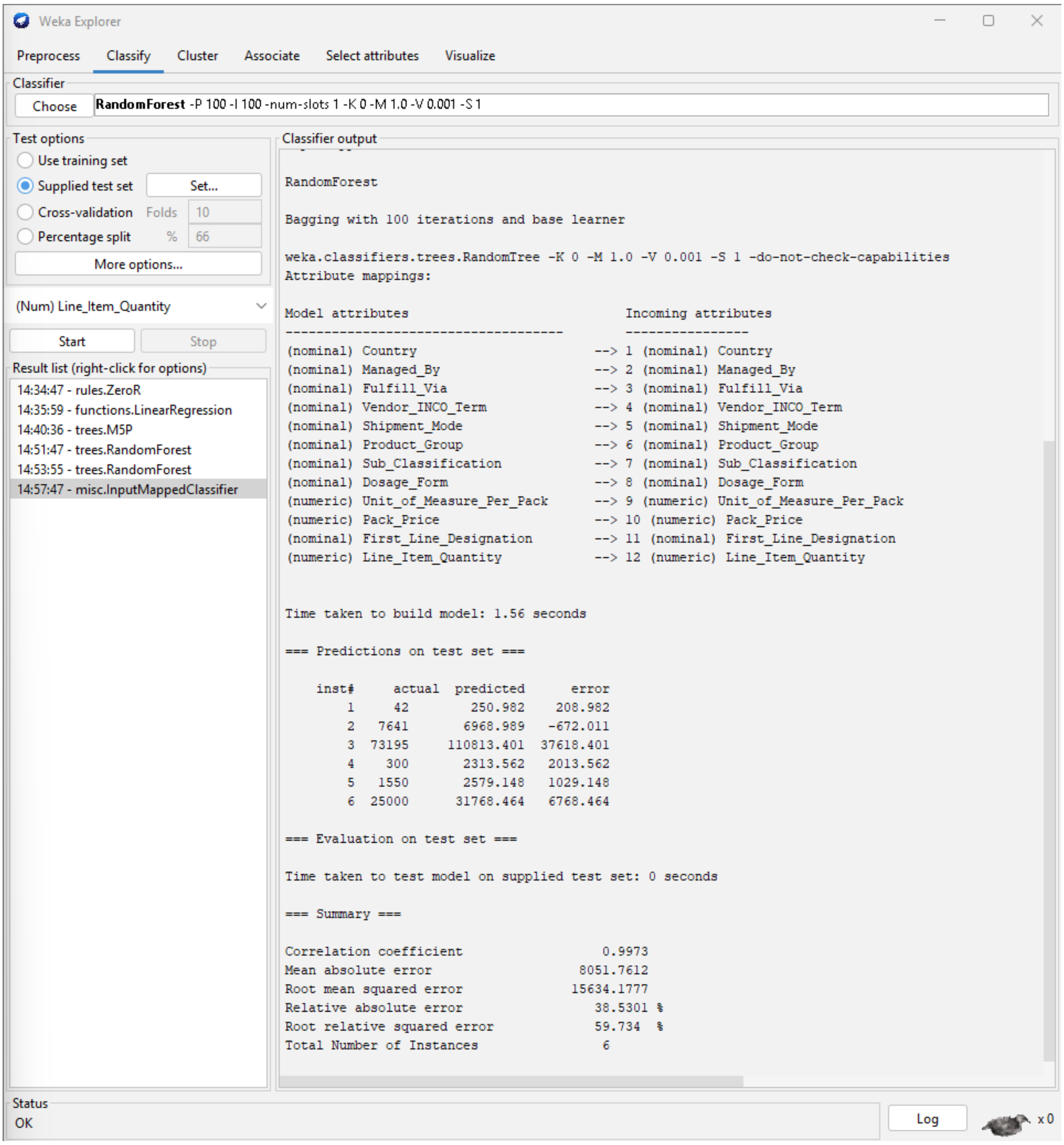

Figure 12: *Validation predictions.*

Figures 11-12 show the WEKA validation setup and predictions. All six instances show the same pattern: the model predicts closer to the training average than the actual quantity. Instance 2 (Mozambique, HRDT, Ocean: actual 7,641, predicted 6,969, -8.8%) is the best result, sitting closest to a typical training shipment. The two largest quantities (Rwanda: 73,195 units; Cote d'Ivoire: 25,000 units) are both over-predicted as the model reverts toward the mean. The two smallest (South Africa: 42

units; Nigeria: 300 units) show the reverse: percentage errors look alarming (+498%, +671%), but absolute gaps are only 209 and 2,014 units. The operationally serious miss is Rwanda at 37,618 units wrong on a 73,195-unit shipment. To mitigate this, two optimisation paths are available: adopting a quantile regression loss function to penalise under-prediction more heavily, or enriching the feature set with site-level patient count and shelf-life constraints currently absent from the proxy data (Studer et al., 2021).

6. Discussion and Transferability

Procedurally valid but substantively incomplete as an LC-CLS foundation.

The ZeroR-to-Random Forest progression satisfies CRISP-ML(Q)'s quality-gate requirement that baseline performance be documented before deploying high-complexity architectures (Studer et al., 2021). Linear Regression and M5P satisfy the interpretability mandate Rudin et al. (2022) identify as non-negotiable in consequential operational settings, codified by EU AI Act Article 9 (Regulation (EU) 2024/1689, 2024). The evaluation metric framework (R, MAE, RMSE, RAE, and RRSE) is directly portable to production. In the LC-CLS context, the asymmetry runs specifically toward under-prediction: a site short of kits risks patient visit delays and GCP protocol deviations, consequences categorically more damaging than the cold-chain waste of surplus inventory (Getz, Smith and Kravet, 2023). This is the operational reason the Unit 9 proposal recommended a quantile regression objective, which assigns differential penalties to under- and over-prediction rather than treating both equally.

Table 3 maps what transfers and what does not: the transferable elements are methodological (leakage detection, distributional inspection, and the CRISP-ML(Q) quality-gate sequence), whilst the performance figures do not. Wornow et al. (2023) identify such structural incompatibilities as more consequential than feature incompleteness alone.

***Table 3:** Proxy-to-production mapping: USAID to LC-CLS*

Dimension	USAID proxy	LC-CLS production
Unit of analysis	Shipment event	Patient visit schedule (per site)
Target variable	Line_Item_Quantity (continuous)	Kit quantity per visit type (V1/V2/Unscheduled)
Core features	Country, product group, shipment mode	Site region, therapeutic area, visit-type kit schedule
Missing critical features	-	Study phase, return probability, shelf-life, site patient count
Data governance	Open access	GDPR Art. 9 special category; consent required
Human approval point	Not modelled	Review gate before allocation confirmed

Assembling LC-CLS patient data engages GDPR Article 9 special category obligations (Regulation (EU) 2016/679, 2016), requiring a consent framework before any dataset can be legitimately assembled. The process, feature engineering, evaluation framework, and CRISP-ML(Q) quality gates are sufficiently validated to carry forward; the performance figures are not.

Kit supply failures across all 43 countries in this dataset (USAID, 2015) reveal the readiness gap; this experiment does not close it.

7. Recommendations

Gradient boosting with a quantile regression objective remains the appropriate production target, directly reducing kit waste and supply shortfalls; the baseline sequence presented here satisfies CRISP-ML(Q)'s quality gate and EU AI Act Article 9's risk management obligations before any higher-complexity deployment (Studer et al., 2021; Regulation (EU) 2024/1689, 2024).

8. Conclusion

This experiment confirms that machine learning can predict healthcare supply chain quantities with meaningful accuracy: Random Forest reduced average error by 43.0% over the naive baseline, consistent with comparable studies (Ganesh and Dass, 2025). The structure, metrics, and CRISP-ML(Q) process transfer to LC-CLS; the error figures do not, pending reconstruction described in Section 6. The remaining errors concentrate in the high-volume tail and very small shipments, which is the expected limitation of this loss function and the reason the quantile regression objective was proposed in Unit 9. Mapping where a prototype falls short is itself a contribution; procedural validity and production readiness are different problems, and this experiment addresses the first. Under CRISP-ML(Q), process and performance are always validated separately; proxy data can confirm the first, never the second.

References

Breiman, L. (2001) 'Random forests', Machine Learning, 45(1), pp. 5-32. Available at: https://doi.org/10.1023/A:1010933404324

Ganesh, L. and Dass, P.L. (2025) 'AI-integrated risk management for healthcare supply chains', Economic Sciences, 21(2), pp. 181-191. Available at: https://doi.org/10.69889/0ve63563

Getz, K., Smith, Z. and Kravet, M. (2023) 'Protocol design and performance benchmarks by phase and by oncology and rare disease subgroups', Therapeutic Innovation and Regulatory Science, 57, pp. 49-56. Available at: https://doi.org/10.1007/s43441-022-00438-5

Kapoor, S. and Narayanan, A. (2023) 'Leakage and the reproducibility crisis in machine-learning-based science', Patterns, 4(9), 100804. Available at: https://doi.org/10.1016/j.patter.2023.100804

Regulation (EU) 2024/1689 (2024) of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union, L 2024/1689, 12 July 2024. Available at: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=OJ:L_202401689 (Accessed: 13 March 2026).

Regulation (EU) 2016/679 (2016) of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation). Official Journal of the European Union, L 119, 4 May 2016, pp. 1-88. Available at: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32016R0679 (Accessed: 13 March 2026).

Rudin, C. et al. (2022) 'Interpretable machine learning: Fundamental principles and 10 grand challenges', Statistics Surveys, 16, pp. 1-85. Available at: https://doi.org/10.1214/21-SS133

Studer, S. et al. (2021) 'Towards CRISP-ML(Q): a machine learning process model with quality assurance methodology', Machine Learning and Knowledge Extraction, 3(2), pp. 392-413. Available at: https://doi.org/10.3390/make3020020

University of Waikato (2024) WEKA 3.8.6: Waikato Environment for Knowledge Analysis. Available at: https://weka.sourceforge.io (Accessed: 13 March 2026).

USAID (2015) Supply Chain Management System (SCMS) Delivery History Dataset. Available at: https://www.kaggle.com/datasets/divyeshardeshana/supply-chain-shipment-pricing-data (Accessed: 13 March 2026).

Wang, W. (2022) 'Root-mean-square error (RMSE) or mean absolute error (MAE): When to use them or not', *Geoscientific Model Development*, 15, pp. 5481-5487. Available at: https://doi.org/10.5194/gmd-15-5481-2022

Wornow, M. et al. (2023) 'The shaky foundations of large language models and foundation models for electronic health records', npj Digital Medicine, 6, p. 135. Available at: https://doi.org/10.1038/s41746-023-00879-8

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Appendix

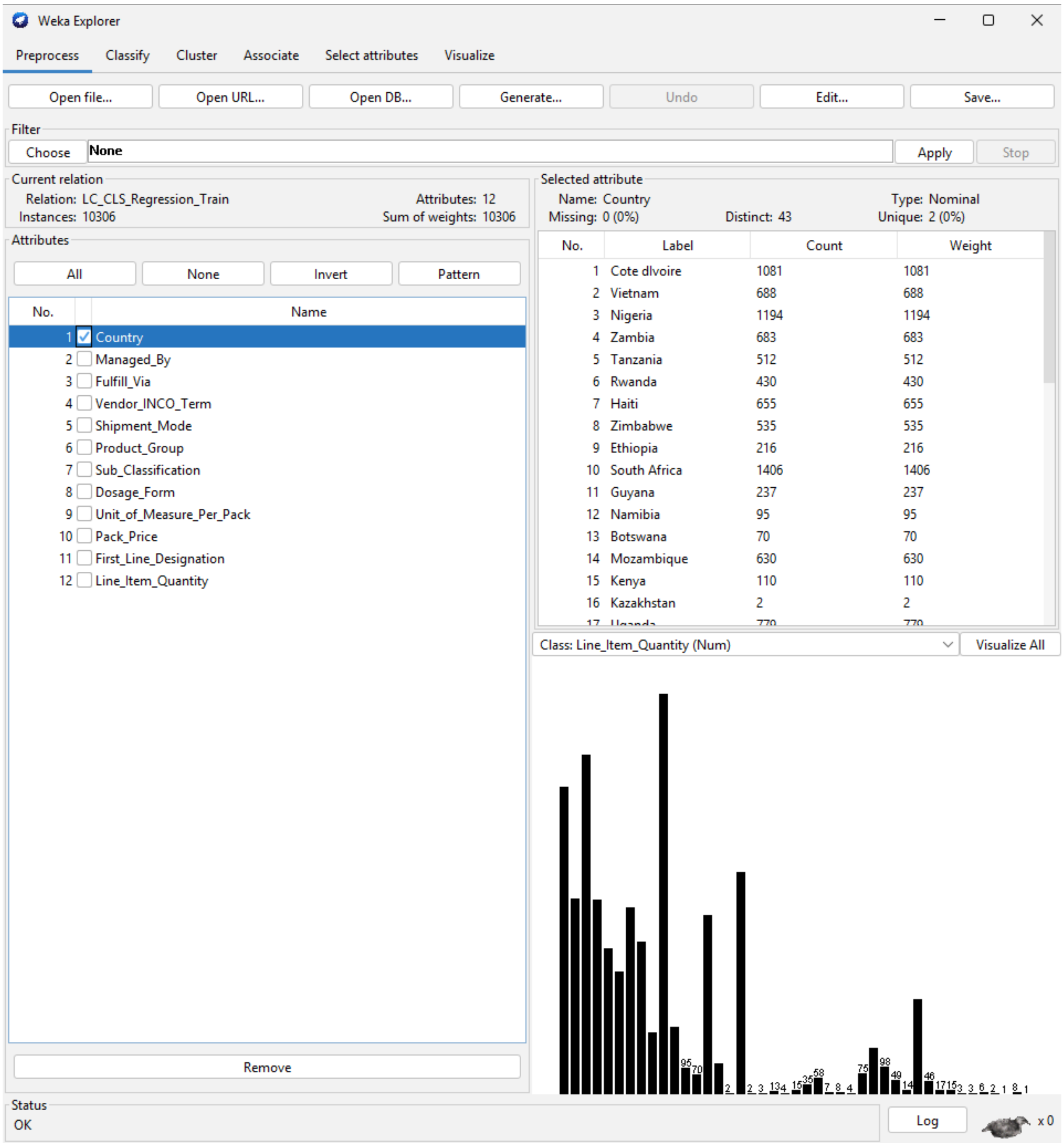

Figure A1: Country - 43 distinct values; South Africa (1,406), Nigeria (1,194), and Cote d'Ivoire (1,081) are the most frequent, reflecting PEPFAR procurement concentration in high-burden HIV countries.

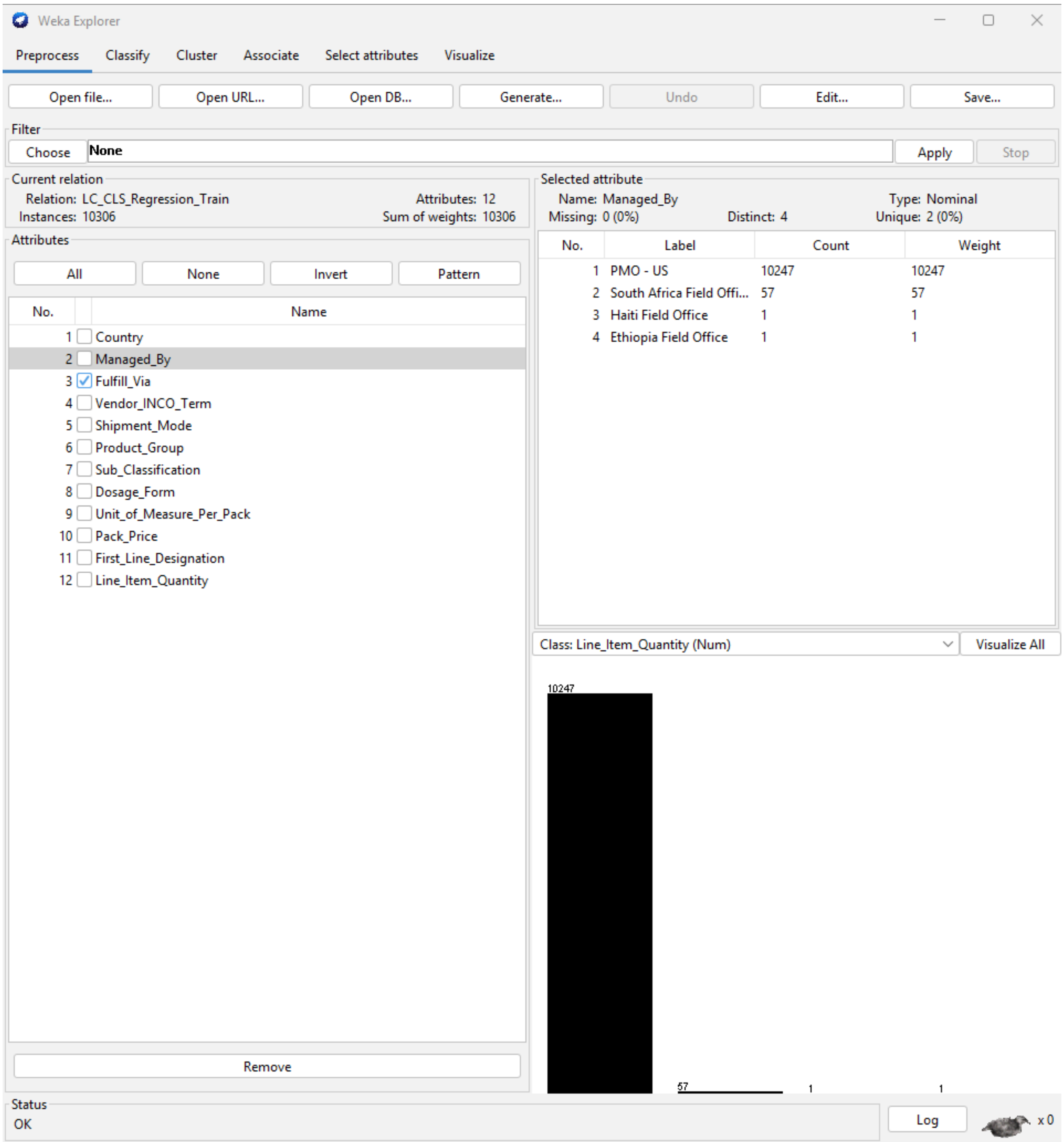

Figure A2: Fulfill_Via - two categories: Direct Drop and From RDC. Reflects the logistical distinction between direct vendor-to-country delivery and regional warehouse routing.

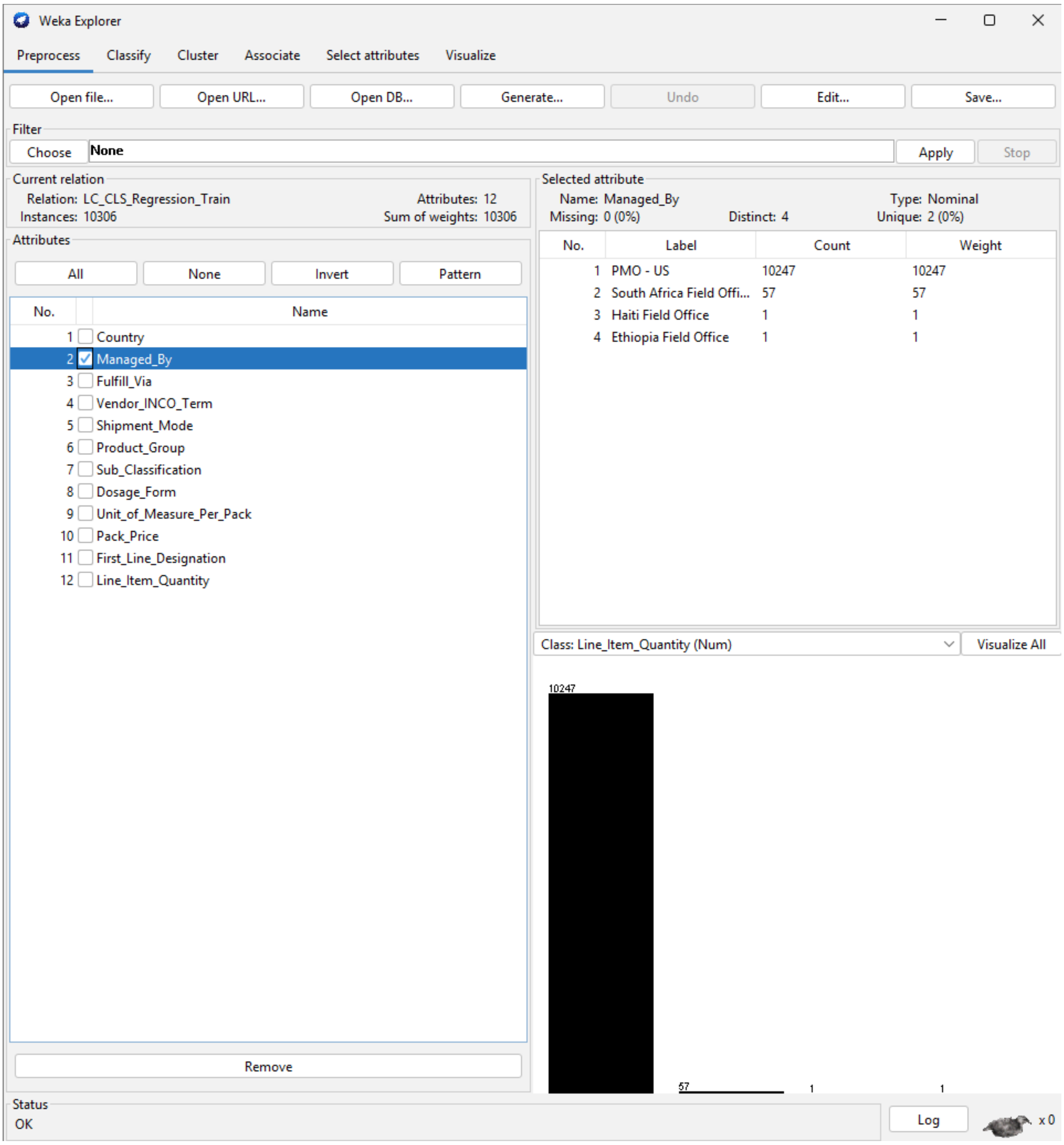

Figure A3: Managed_By - four categories; PMO-US dominates at 99.4% (10,247 rows). Near-zero variance means this attribute contributes minimal discriminative information but is retained for completeness.

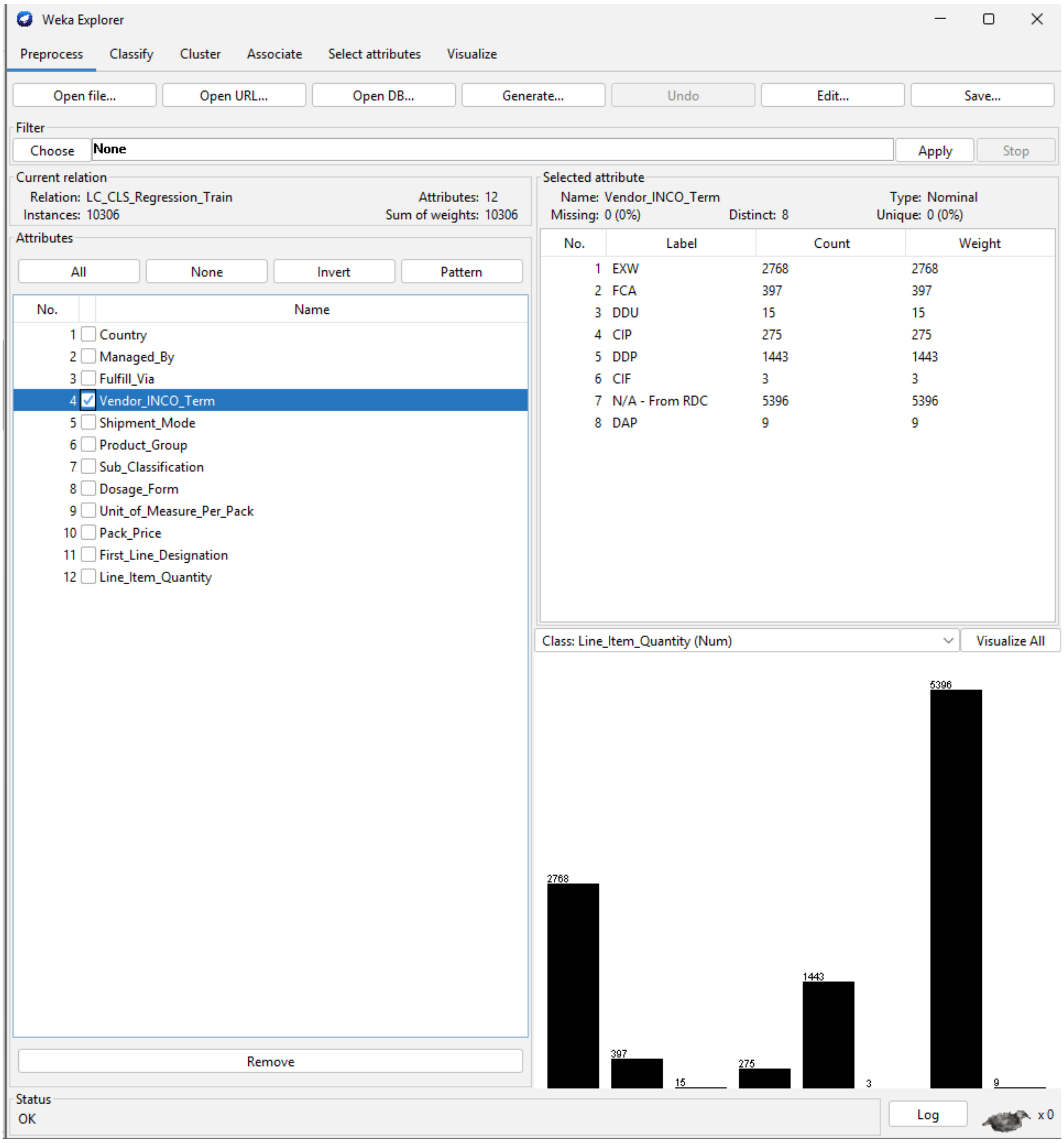

Figure A4: Vendor_INCO_Term - eight categories; N/A-From RDC (5,396) and EXW (2,768) dominate. INCO terms encode contractual handover points and associated logistical complexity.

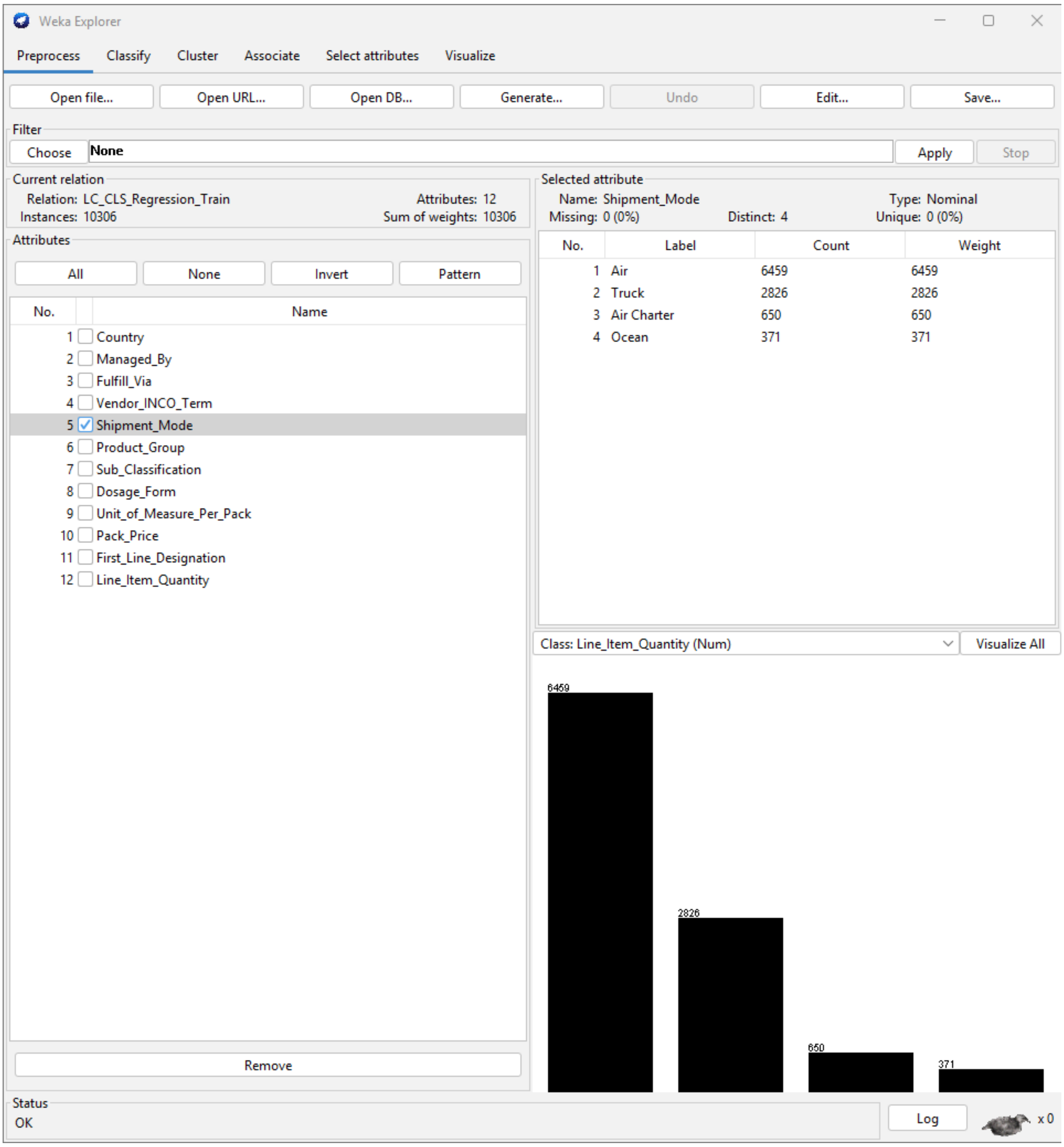

Figure A5: Shipment_Mode - four categories; Air (6,459, 62.7%) predominates, consistent with the urgency tier of antiretroviral procurement. Ocean (371) is reserved for bulk, long-haul deliveries.

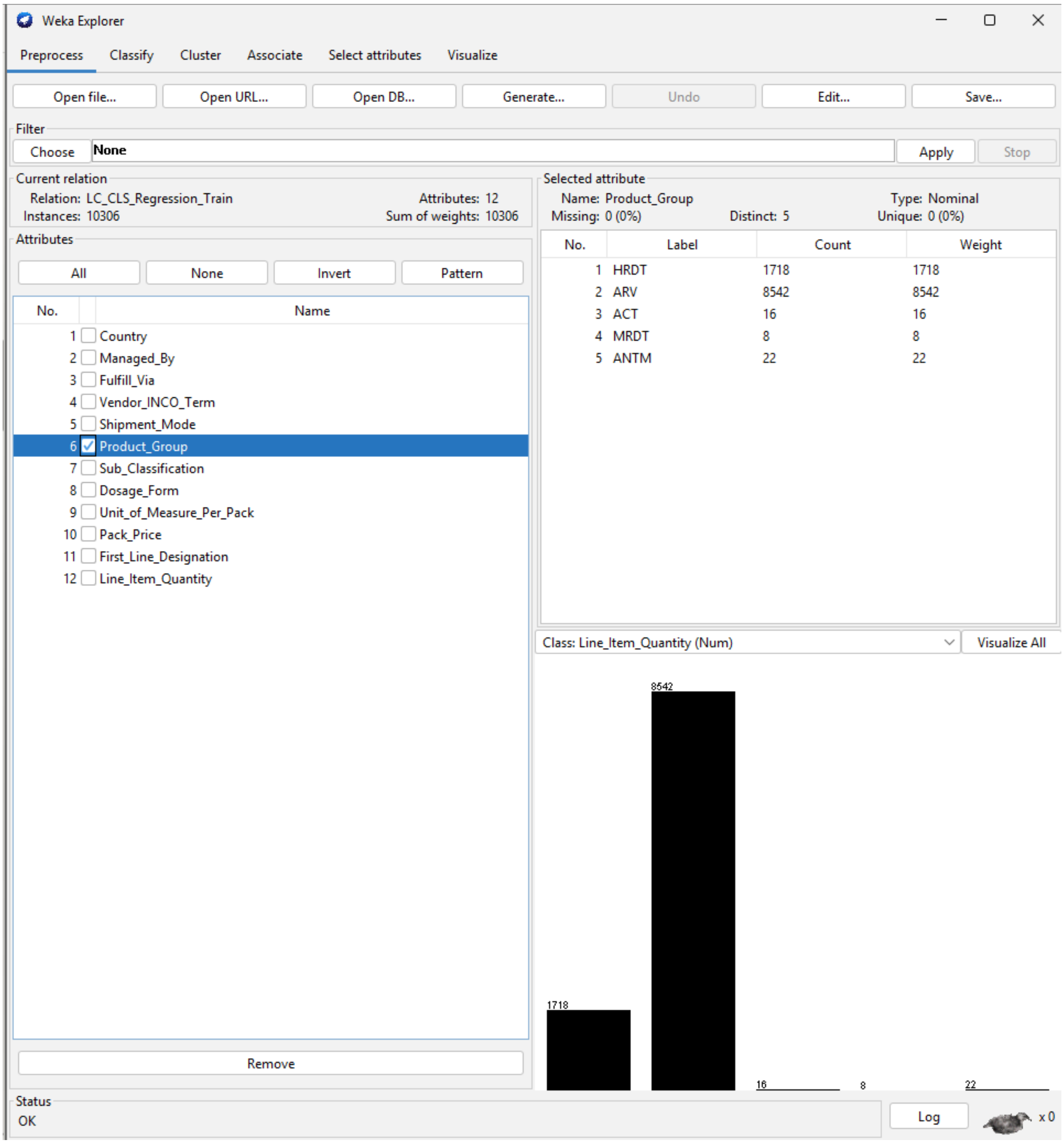

Figure A6: Product_Group - five categories; ARV dominates at 82.9% (8,542 rows). Regression performance primarily reflects ARV procurement dynamics; ACT, MRDT, and ANTM predictions should be interpreted cautiously given sparse training representation.

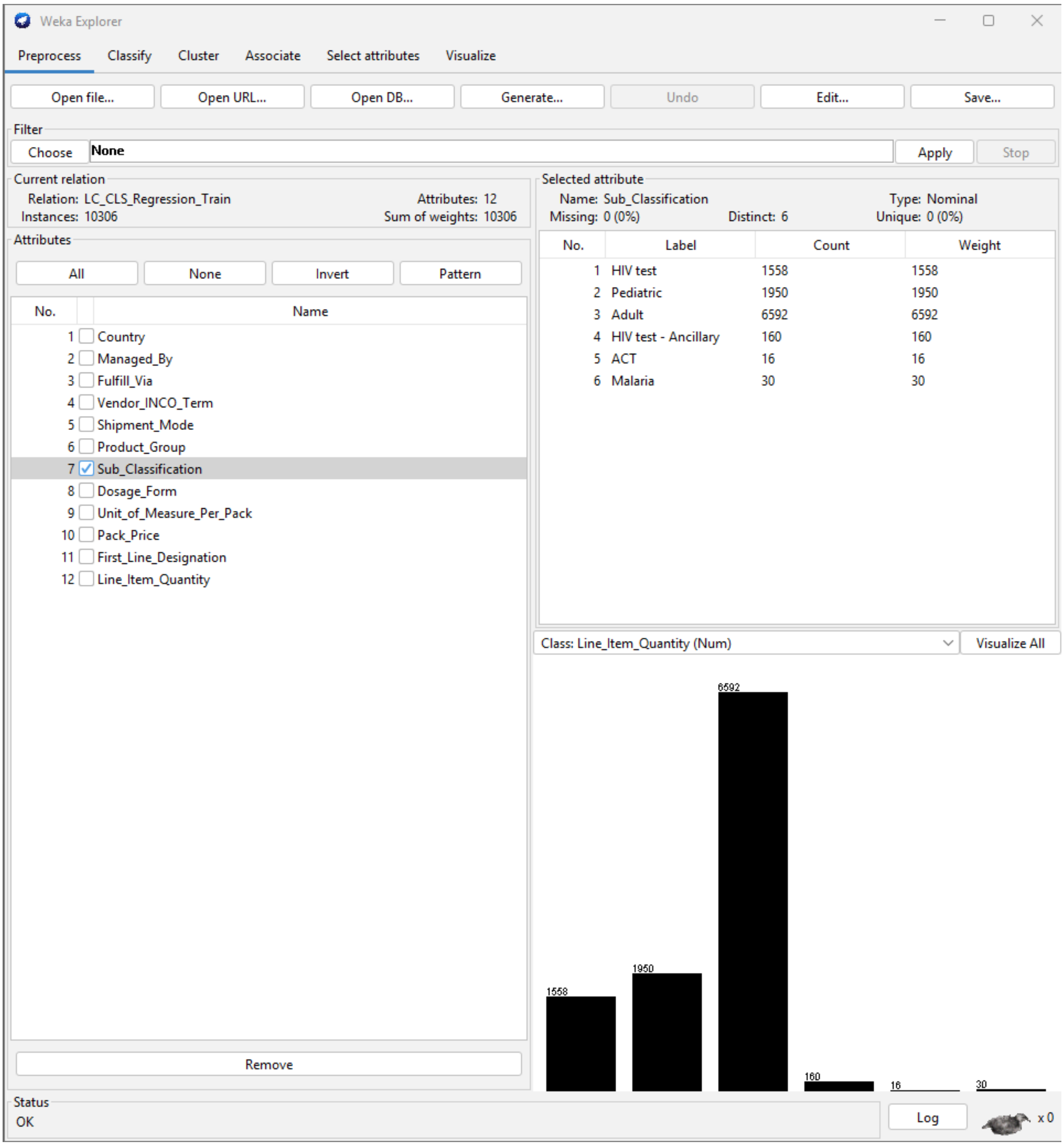

Figure A7: Sub_Classification - six categories; Adult (6,592) and Pediatric (1,950) capture the ARV sub-population. Adult formulations are typically ordered in larger quantities.

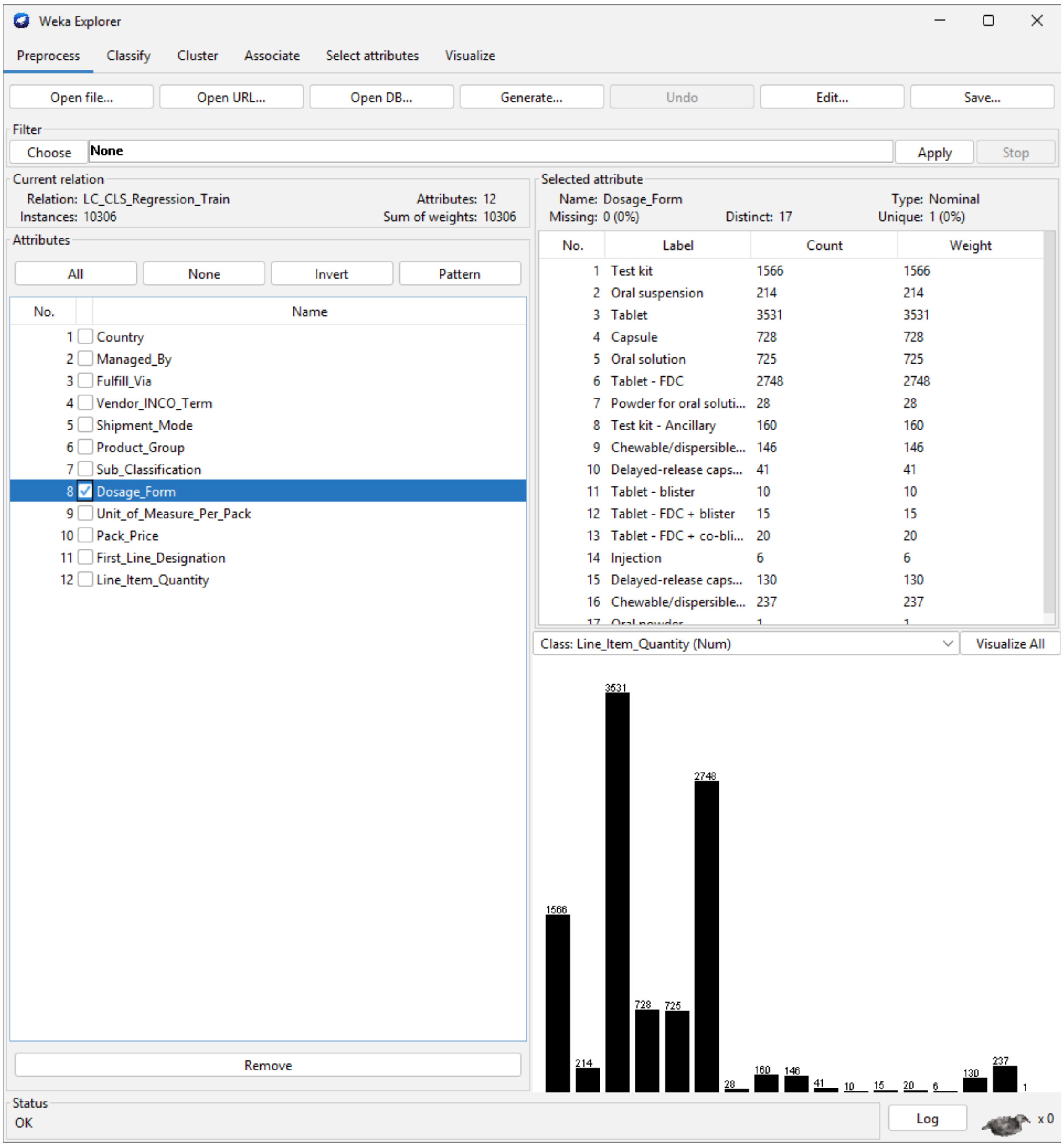

Figure A8: Dosage_Form - 17 distinct categories; Tablet (3,531) and Tablet-FDC (2,748) are the most frequent. High cardinality increases model complexity, particularly in tree-based approaches.

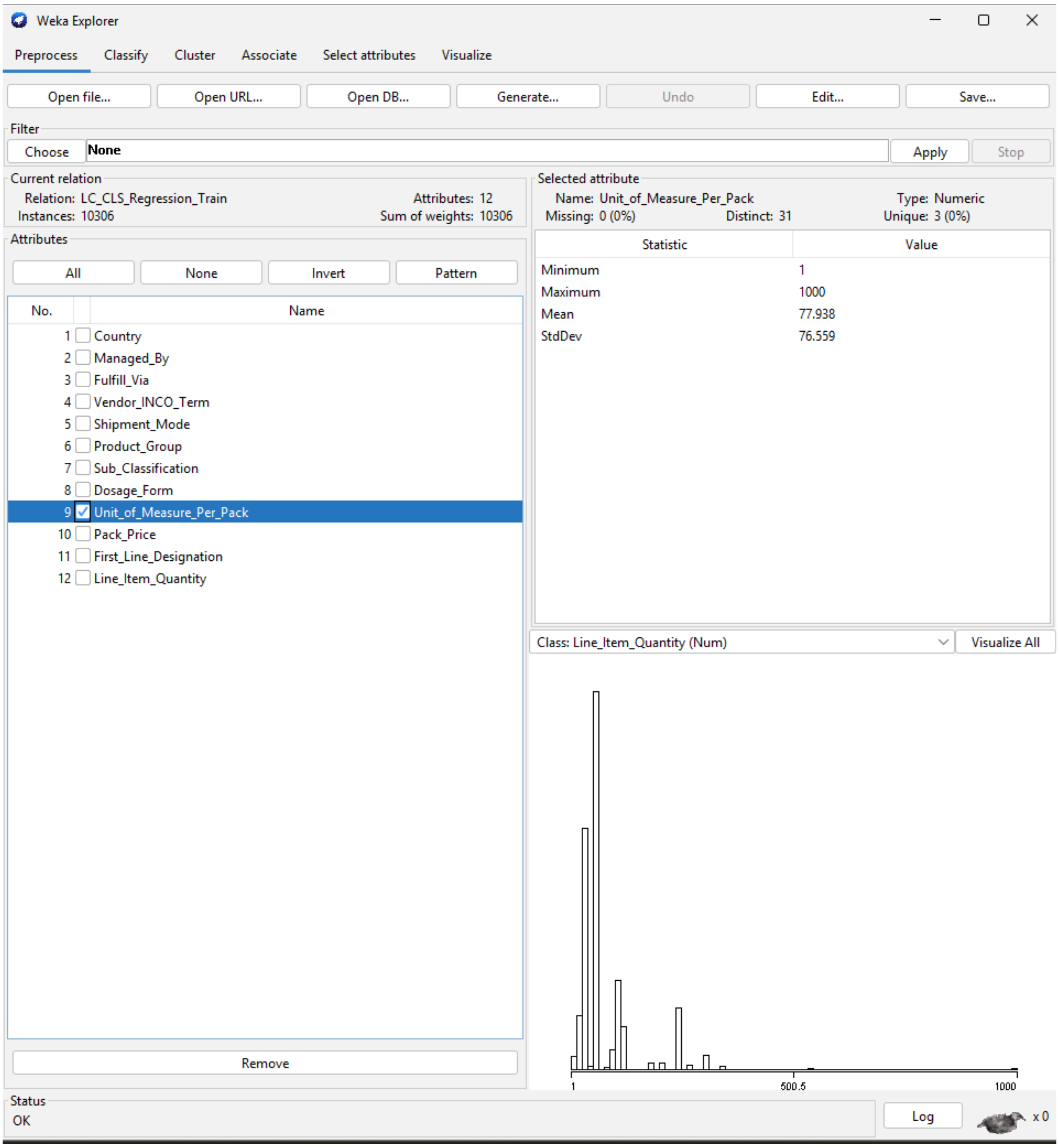

Figure A9: Unit_of_Measure_Per_Pack - numeric, range 1-1,000 (mean 77.9, std 76.6). Concentration at standard dispensing sizes (30, 60, 90 units) with a right-skewed tail for bulk packaging.

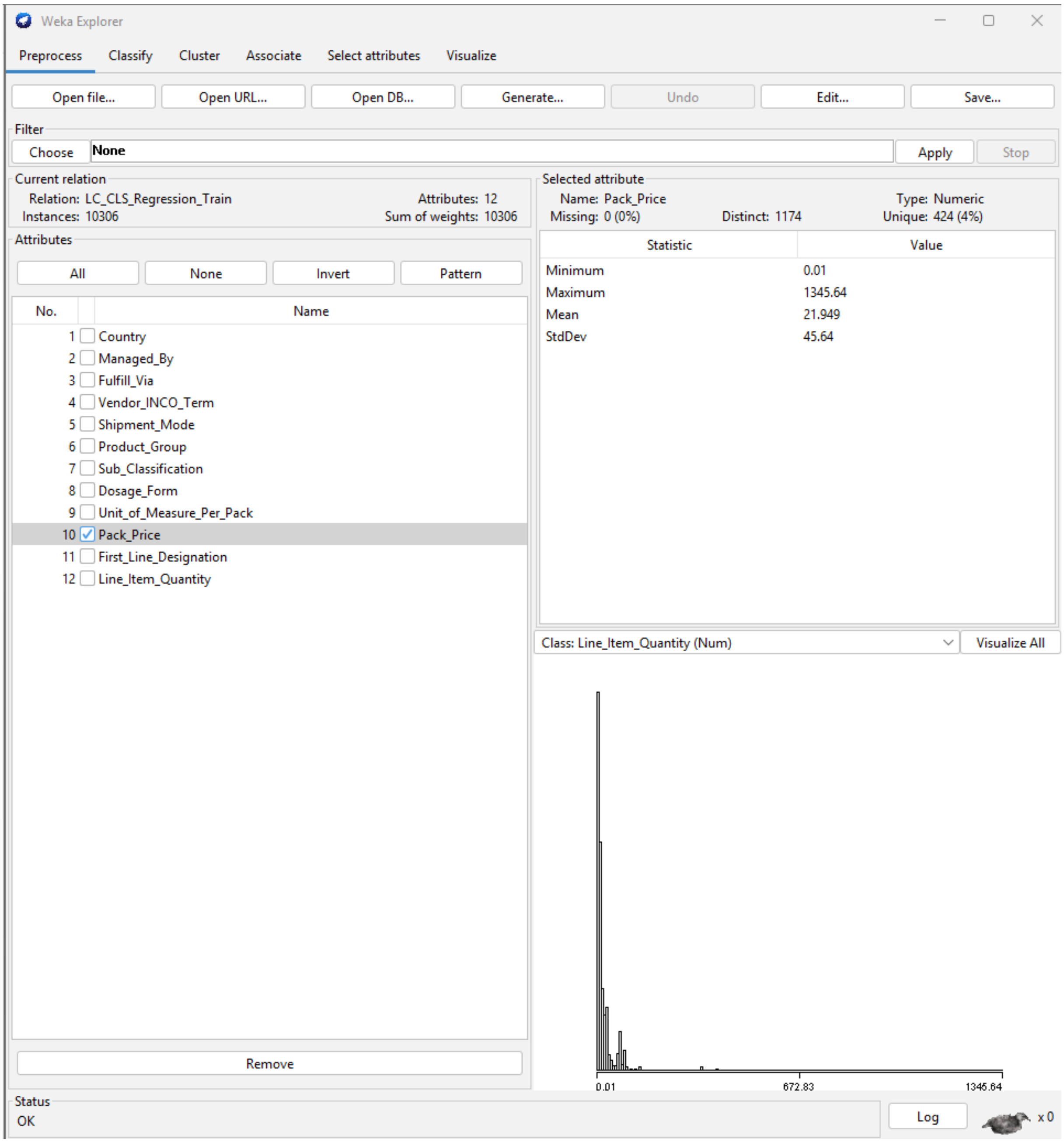

Figure A10: Pack_Price - numeric, range 0.01-1,345.64 (mean 21.95, std 45.64). Substantially right-skewed, reflecting the price differential between inexpensive generic ARVs and specialist biologics.

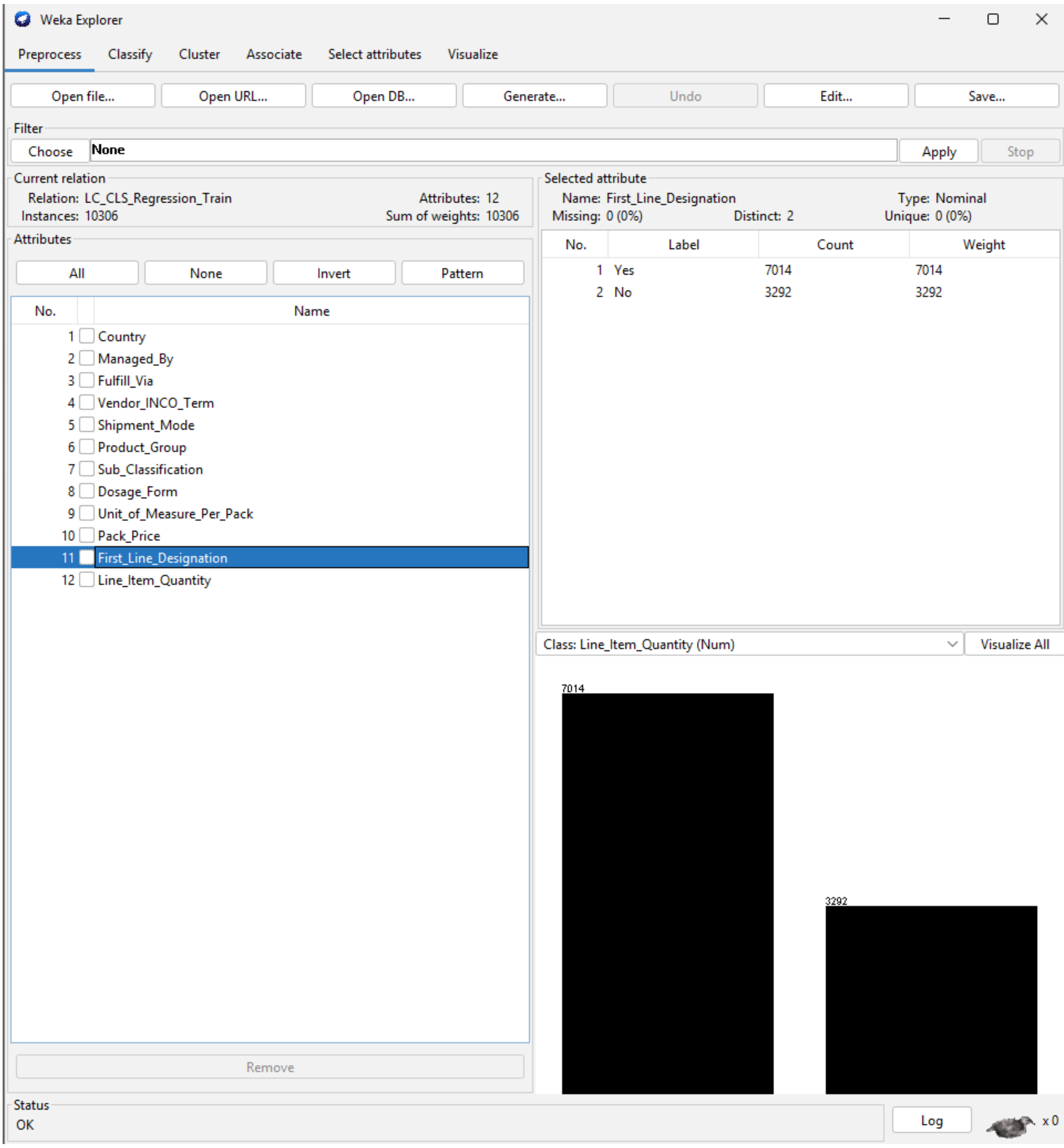

Figure A11: First_Line_Designation - binary; Yes (7,014, 68%) and No (3,292). First-line therapies are procured in substantially higher volumes, making this an important predictor.